

# **KCD New York**

## **2025**



# Empowering Cloud Native Development: Driving Data-Driven Decisions with OpenTelemetry

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# Ronny Kohavi, Microsoft



**“Only 1 out of 3 features improves key metrics.”**

**Most software today ships based on intuition or deadlines.**

# The Problem



**10%** are implementing **Fullstack Observability.** <sup>1</sup>

**40%** say **complexity and cost** are the principal reasons.

<sup>2</sup>

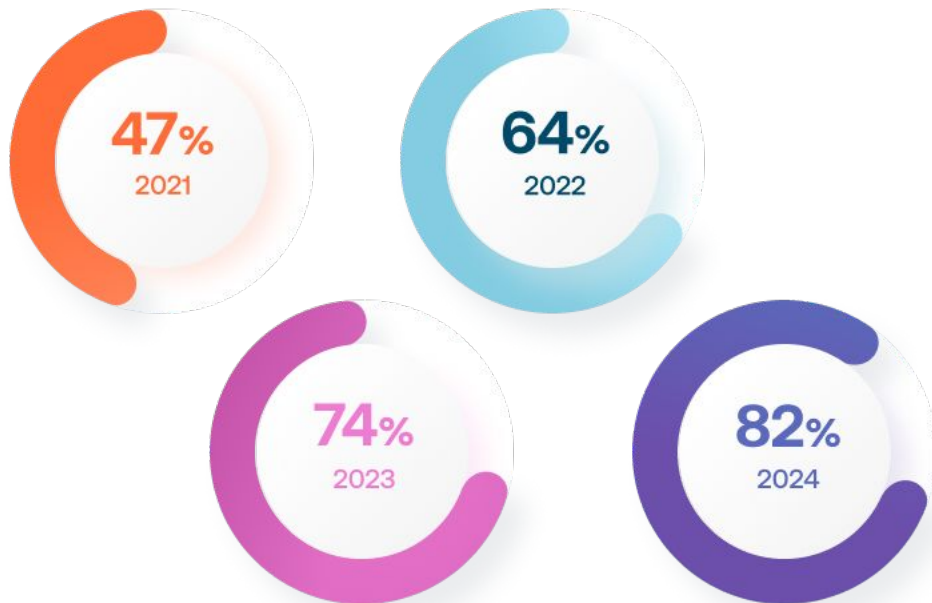
**97% report barriers** to fully investments <sup>3</sup>

1. 2024 Observability Pulse Report By Logz.io

2. Observability Survey 2025 from Grafana

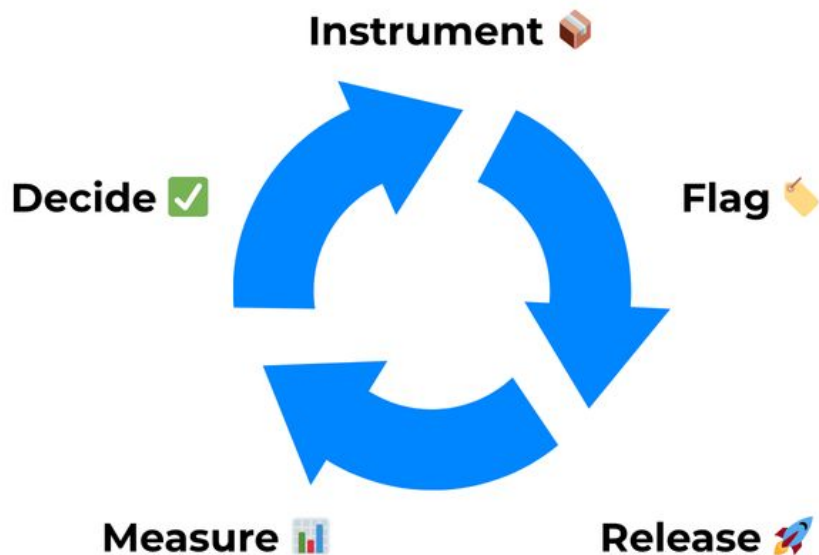
3. The landscape of observability in 2025 from Elastic

# The Scary Trend



**MTTR is on the rise for  
the 3rd consecutive year**

# The Experiment Loop



The loop enables teams to:

- ✓ Experiment safely
- ✓ Remediate faster
- ✓ Improve continuously

# Build your tool chain

Instrument 🕒

Flag 🚩

Release 🚀

Measure 📐

Decide 👤



Collect metrics & traces from services



flagger

Rollout features safely

Open  
Feature



Enable modern  
deployment  
strategies



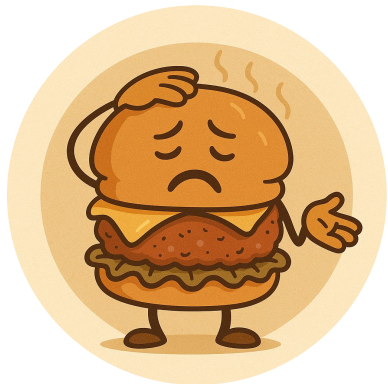
Track KPIs and  
SLAs



Visualization, Alerts,  
Diagnose

AI & LLM  
Grafana Sift

# Measure what matters to the customer



**Orders Returned**



**Order Preparation Latency**



**Order Accuracy Rate**





# Start small



If it can wake you at 3 AM, it must be monitored.

# Measure what matters to each department

## Business

 **Revenue**

 **% Feature adoption**

 **% Customer growth**

## Dev & Ops

 **MTTD**

 **MTTR**

 **Service availability**

# Find your business champion



**Someone that could  
bridge between **business**  
and **engineering****

# Promote your PoC to your champion!

 Open  
Feature

 Marketing

 A/B Testing



 Sales

 % Feature adoption



 Security

 Reduce security  
incidents

# Generate Failed Customer Interaction (FCI)

```
/20 07:52:26 /pay request from 172.24.0.5:44678
/20 07:52:26 x payment failed for {Amount:42 Currency:USD CardNumber:4111111111111111 ExpiryMMYY:12/25 CVV:123}: payment declined by issuing bank
/20 07:52:26 /pay request from 172.24.0.5:44678
/20 07:52:26 x payment failed for {Amount:42 Currency:USD CardNumber:4111111111111111 ExpiryMMYY:12/25 CVV:123}: payment declined by issuing bank
/20 07:52:26 /pay request from 172.24.0.5:44694
/20 07:52:26 ✓ payment succeeded for {Amount:42 Currency:USD CardNumber:4111111111111111 ExpiryMMYY:12/25 CVV:123}
/20 07:52:26 /pay request from 172.24.0.5:44708
/20 07:52:26 ✓ payment succeeded for {Amount:42 Currency:USD CardNumber:4111111111111111 ExpiryMMYY:12/25 CVV:123}
```

$$FCI = \frac{\text{n. of interactions failed}}{\text{total of interactions failed}}$$

$$100 * \frac{\text{sum(rate(customer_interactions_failed_total[5m]))}}{\text{sum(rate(customer_interactions_total[5m]))}}$$



# Fullstack Observability



Business



User Experience



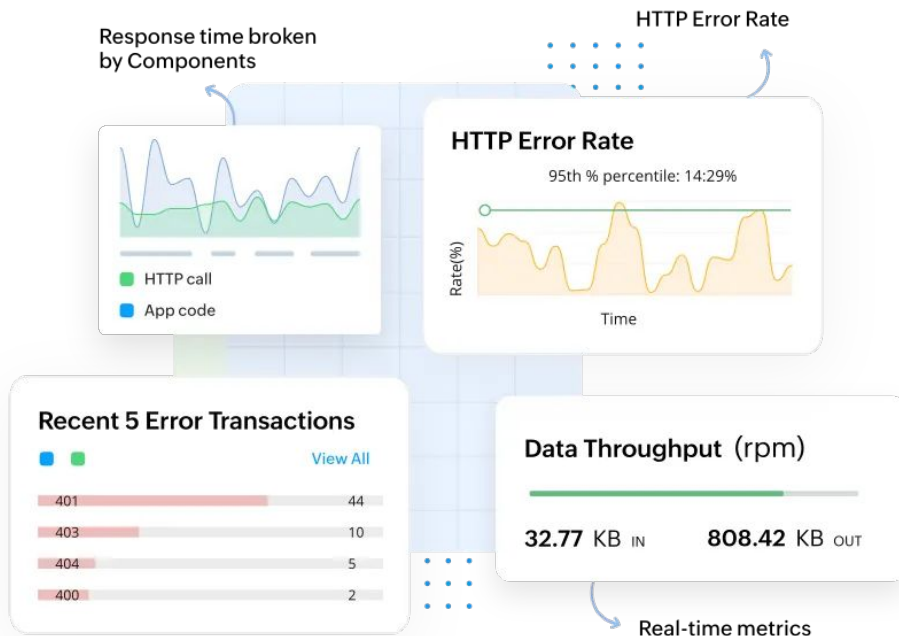
Runtime Security



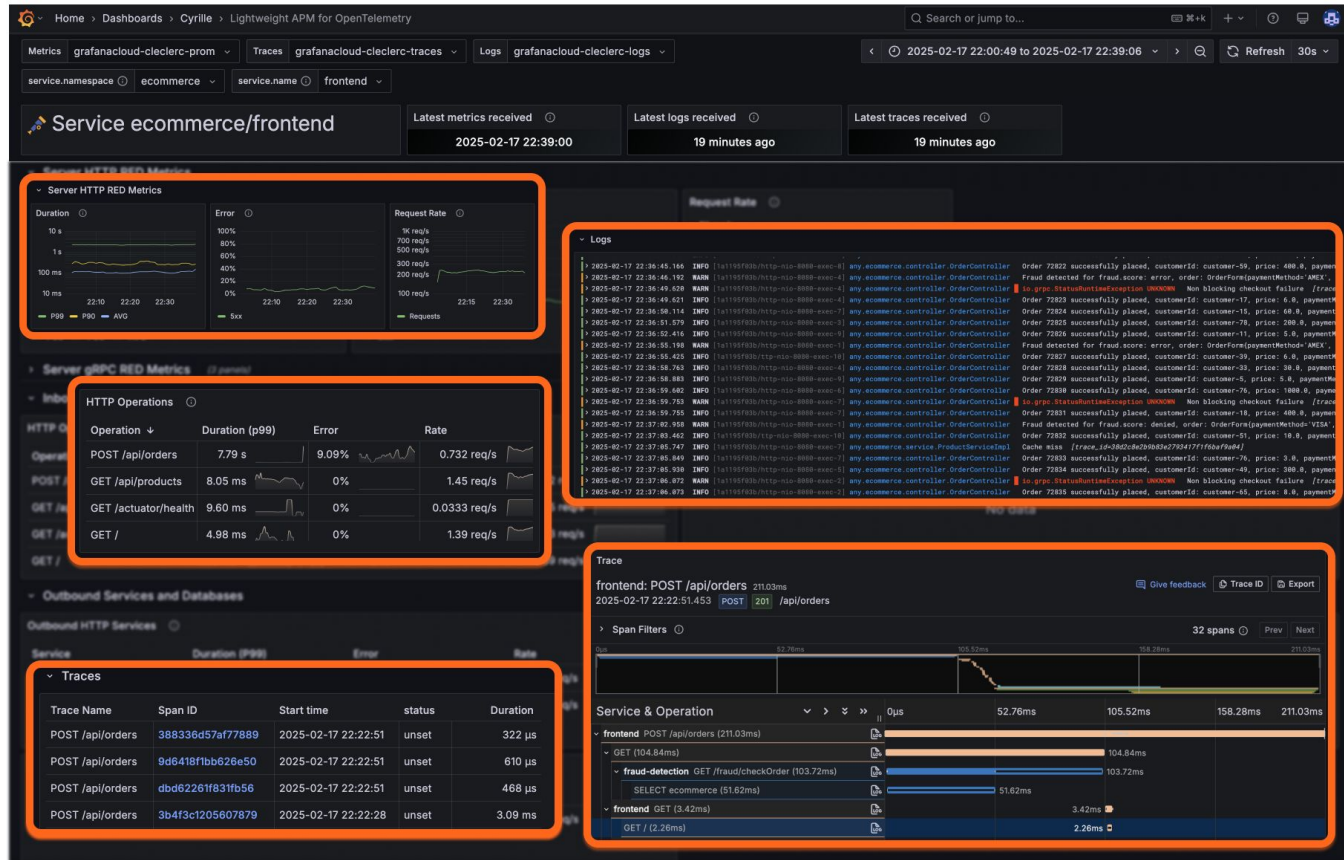
Application



Infrastructure



# APM Dashboard



# Is OpenTelemetry the definitive solution?



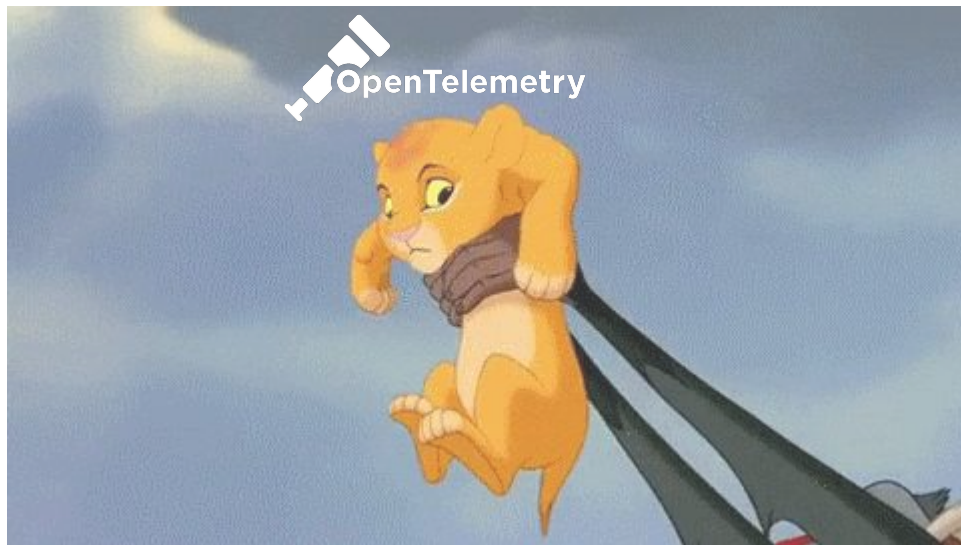
Growing ecosystem



Vendor-neutral



**Language and feature support**



<https://opentelemetry.io/status/>



# Tech is just a tool

It always begins with a business problem

# Thanks NYC



# Resources

- <https://opentelemetry.io/status/>
- <https://opentelemetry.io/docs/languages/>
- **Unlocking Customer-Centric Observability, Kokilavani Kathiresan Observability Day 2025**
- <https://grafana.com/grafana/dashboards/19419-opentelemetry-apm/>
- **Operationalizing Observability, Hazel Weakly, Observability Day 2025**

